

Binomial distributions

AB 514

Repeated trials:

A random experiment with only two possible outcomes namely success & failure is called a Bernoulli trial. Where the probability of success p is same for each trial.

Binomial distribution:

If p is the probability of success & q is the probability of failure, then the probability of ' x ' success out of ' n ' trials is given by

$$P(x) = {}^nC_x p^x q^{n-x} \quad ; \text{ it is also denoted by}$$

$$B(x; n, p) \text{ @ } B(n, p)$$

$x \rightarrow \text{Binomial variable}$

Where n & p are the parameters & $x = 0, 1, 2, \dots, n$

The Binomial distribution with parameters n & p can also be written as follows

$$\begin{array}{ccccccc} x : & 0 & 1 & 2 & \dots & \dots & n \\ P(x) : & q^n & n_1 p q^{n-1} & n_2 p^2 q^{n-2} & \dots & \dots & p^n \end{array}$$

Here the value of $P(x)$ for different values $x = 0, 1, 2, \dots, n$ are the successive terms in the binomial expansion of $(q+p)^n$ & this distribution is called the Binomial distribution @ Bernoulli distribution.

$$\text{Now } \sum P(x) = q^n + n_1 p q^{n-1} + n_2 p^2 q^{n-2} + \dots + p^n = (p+q)^n = 1^n = 1$$

Hence $P(x)$ is a probability function

② Bernoulli distribution (James Bernoulli)

Binomial distribution

If p is the probability of success & q is the Probability of failure, then the probability of x success out of n trials is given by

$$P(x) = {}^nC_x p^x q^{n-x} = B(n, p)$$

Where n, p are the parameters & $x = 0, 1, 2, \dots, n$

Here ${}^nC_x p^x q^{n-x}$ is the x^{th} term of the binomial expansion $(q+p)^n$, this distribution is called the binomial distribution. & $P(x)$ is called the Probability function.

Now

$$\sum P(x) = q^n + {}^nC_1 p q^{n-1} + {}^nC_2 p^2 q^{n-2} + \dots + p^n = (q+p)^n = (1)^n = 1$$

The random variable x associated with a binomial distribution is called a binomial Variate.

Binomial Distribution is applicable, when

- ① Probability of success remains same for every trial
- ② Each trial should be independent
- ③

* Mean & S.D of the Binomial distribution

$$\text{WKT mean } (\mu) = \sum_{x=0}^n x p(x)$$

$$= \sum_{x=0}^n x {}^n C_x P^x q^{n-x}$$

$$= \sum_{x=0}^n x \cdot \frac{n!}{x! (n-x)!} \cdot P^x \cdot q^{n-x}$$

$$= \sum_{x=1}^n \frac{n(n-1)!}{(x-1)! (n-x)!} P \cdot P^{x-1} \cdot q^{n-x}$$

$$= nP \sum_{x=1}^n \frac{(n-1)!}{(x-1)! [(n-1)-(x-1)]!} P^{x-1} q^{(n-1)-(x-1)}$$

$$= nP \sum_{x=1}^n {}^{(n-1)}C_{(x-1)} P^{x-1} q^{(n-1)-(x-1)}$$

$$= nP \left[{}^{(n-1)}C_0 P^0 q^{n-1} + {}^{(n-1)}C_1 P^1 q^{n-2} + \dots + {}^{(n-1)}C_{n-1} P^{n-1} q^0 \right]$$

$$= nP (P+q)^{n-1}$$

$$= nP (1)^{n-1}$$

$$\boxed{\mu = np}$$

Variance $V(x) = E(x^2) - [E(x)]^2$

$$= \sum_{x=0}^n x^2 p(x) - \mu^2$$

$$= \sum_{x=0}^n x^2 p(x) - (np)^2 \rightarrow \textcircled{1}$$

Consider $\sum_{x=0}^n x^2 p(x) = \sum_{x=0}^n [x(x-1) + x] p(x)$

$$= \sum_{x=0}^n x(x-1) p(x) + \sum_{x=0}^n x p(x)$$

$$= \sum_{x=0}^n x(x-1) {}^n C_x P^x q^{n-x} + \sum_{x=0}^n x p(x)$$

$$= \sum_{x=0}^n x(x-1) \frac{n!}{x!(n-x)!} p^x q^{n-x} + np$$

$$= \sum_{x=0}^n x(x-1) \frac{n(n-1)(n-2)!}{x(x-1)(x-2)!(n-x)!} p^{x-2} p^2 q^{n-x} + np$$

$$= \sum_{x=2}^n \frac{n(n-1)(n-2)!}{(x-2)!(n-x)!} \cdot p^{x-2} p^2 q^{n-x} + np$$

$$= n(n-1)p^2 \sum_{x=2}^n \frac{(n-2)!}{(x-2)!(n-x)!} p^{x-2} q^{n-x} + np$$

$$= n(n-1)p^2 \sum_{x=2}^n \binom{n-2}{x-2} p^{x-2} q^{n-x} + np$$

$$= n(n-1)p^2 \left[\binom{n-2}{0} p^0 q^{n-2} + \binom{n-2}{1} p^1 q^{n-3} + \dots + \binom{n-2}{n-2} p^{n-2} q^0 \right] + np$$

$$= n(n-1)p^2 (q+p)^{n-2} + np$$

$$= n(n-1)p^2 + np$$

Substitute this in (1), we get

$$V(x) = n(n-1)p^2 + np - np^2$$

$$= \cancel{np^2} - np^2 + np - \cancel{np^2}$$

$$= np - np^2$$

$$= np(1-p)$$

$$= npq$$

$$\boxed{V(x) = npq}$$

$$\therefore \text{S.D.} = \sqrt{V(x)}$$

Problems

- (1) let x be a binomially distributed random Variable with mean 2 & s.d. $2/\sqrt{3}$. find the corresponding probability function

Soln:- Given $\mu = 2$, $\sigma = \frac{2}{\sqrt{3}}$

we have $\mu = np$ & $\sigma = \sqrt{npq}$

$$2 = np \quad \& \quad \frac{2}{\sqrt{3}} = \sqrt{npq} \Rightarrow npq = \frac{4}{3}$$

$$np = 2 \rightarrow (1) \quad np = \frac{4}{3q} \rightarrow (2)$$

$$\therefore q = \frac{4^2}{3q^2}$$

$$q = \frac{2}{3}$$

$$p = 1 - q = 1 - \frac{2}{3} \Rightarrow p = \frac{1}{3}$$

from (1) $n \cdot \frac{1}{3} = 2 \Rightarrow n = 6$

The binomial probability fcn $p(x) = {}^nC_x p^x q^{n-x}$

$$\Rightarrow p(x) = {}^6C_x \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{6-x}$$

- (2) When a Coin is tossed 4 times, find the Probability of getting (i) exactly one head (ii) at most 3 heads & (iii) at least two heads

Soln:- let probability of getting head as success

$$\text{i.e. } P(H) = \frac{1}{2} = p \Rightarrow P(T) = \frac{1}{2} \Rightarrow q = 1 - p = \frac{1}{2} \quad \& \quad n = 4$$

WKT the probability of x success out of n trials is given by,

$$P(x) = {}^nC_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x}$$

$$(i) P(\text{one head}) = P(x=1) = {}^4C_1 \left(\frac{1}{2}\right) \left(\frac{1}{2}\right)^{4-1} = 0.25$$

$$(ii) P(\text{at most 3 heads}) = P(x=0) + P(x=1) + P(x=2) + P(x=3)$$

$$= 4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4 + 4C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3 + 4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 + 4C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^1 + 4C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^0$$

$$= \underline{0.9375}$$

(iii) $P(\text{at least 2 heads}) = P(X=2) + P(X=3) + P(X=4)$

$$= 4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 + 4C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^1 + 4C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^0$$

$$= \underline{0.6875}$$

3) The probability that a man aged 60 will live upto 70 is 0.65. What is the probability that out of 10 men, now aged 60, at least 7 will live upto 70

Soln:- let x be the no of persons aged 60 years living upto 70 years

G.T $P = 0.65$, $q = 1 - p = 1 - 0.65 = 0.35$ & $n = 10$.

We have $P(x) = {}^{10}C_x (0.65)^x (0.35)^{10-x}$

We have to find

$$P(X \geq 7) = P(X=7) + P(X=8) + P(X=9) + P(X=10)$$

$$= \underline{0.5138}$$

4) An airline knows that 5% of people making reservations on a certain flight will not turn up. Consequently, their policy is to sell 52 tickets for a flight that can only hold 50 passengers. What is the probability that there will be a seat available for every passenger who turns up?

Soln:- let x denote the no of passengers who will not turn up
for variate, we have

$$P = \frac{5}{100} = 0.05, \quad \therefore q = 1 - 0.05 = 0.95 \quad \& \quad n = 52$$

$$\therefore P(x) = {}^nC_x p^x q^{n-x} = {}^{52}C_x (0.05)^x (0.95)^{52-x}$$

A seat is assured for every passenger who turns up. If the no of passenger who fail to turn up is more than 2 equal to 2. There will be a seat for every passenger who

$$\begin{aligned}\therefore P(x \geq 2) &= 1 - [P(x < 2)] \\ &= 1 - [P(x=0) + P(x=1)] \\ &= 1 - [(0.95)^{52} + {}^{52}C_1 (0.05)(0.95)^{51}] \\ &= 1 - 0.2595 \\ &= 0.7405\end{aligned}$$

$\therefore$ Probability that a seat is available for every passenger is 0.7405

(5) The no of telephone lines busy at an instant of time is a binomial variate with probability 0.2. If at an instant 10 lines are chosen at random, what is the probability that (i) 5 lines are busy? (ii) at most 2 lines are busy

Soln:- let x denotes the no of telephone lines busy. for this variate

we have $p=0.2$, $q=0.8$, $n=10$

Now the R.D is $P(x) = {}^{10}C_x (0.2)^x (0.8)^{10-x}$

$$\begin{aligned}\text{(i) Prob. of 5 lines are busy} &= P(x=5) = {}^{10}C_5 (0.2)^5 (0.8)^5 \\ &= \end{aligned}$$

$$\begin{aligned}\text{(ii) Prob. at most 2 lines are busy} &= P(x=0) + P(x=1) + P(x=2) \\ &= \end{aligned}$$

Determine the probability of getting ~~4~~9 exactly twice in 3 throws with a pair of fair dice

Ans:- Prob. of getting 9 when a pair of dice is thrown $\rightarrow \{ (6,3) (3,6) (5,4) (4,5) \}$

$$\text{Prob. of getting 9} = \frac{4}{36} = \frac{1}{9}$$

$$n=3$$

Prob of getting 9 exactly Twice in 3

$$\text{Throws} = {}^nC_x p^x q^{n-x}$$

$$= {}^3C_2 \left(\frac{1}{9}\right)^2 \left(\frac{8}{9}\right)^{3-2}$$

$$= \frac{8}{243}$$

Out of 800 families with 5 children each, how many would you expect to have 3 boys (b) 5 girls (c) either 2 or 3 boys
Assume equal prob. for boys & girls

$$P(B) = \frac{1}{2} = p$$

$$P(G) = q = \frac{1}{2}$$

Ans:- $n=5$; $X \rightarrow$ no of ^{boys} ~~families~~ in the family

$${}^5C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x}$$

$$x=3) = {}^5C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = \frac{5}{16}$$

800 families, expected no of families have boys out of 5 childrens $\Rightarrow 800 \times \frac{5}{16} = \underline{250}$

$$\textcircled{b} \quad P(X=0) = \left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

$\therefore 800 \times \frac{1}{32} = 25$ families have 5 girls

$$\textcircled{c} \quad P(X=2 \text{ or } X=3) = P(2) + P(3)$$

∴ let x be the probability of defective pens

$$\therefore p = 0.1, \quad q = 1 - p = 1 - 0.1 = 0.9, \quad n = 12$$

$$\therefore P(x) = {}^{12}C_x (0.1)^x (0.9)^{12-x}$$

we need to find

$$P(x=2) = {}^{12}C_2 (0.1)^2 (0.9)^{12-2} = 66 \times 0.01 \times 0.2824 = \underline{0.2001}$$

$$\begin{aligned} P(x \leq 2) &= P(x=0) + P(x=1) + P(x=2) \\ &= {}^{12}C_0 (0.1)^0 (0.9)^{12} + {}^{12}C_1 (0.1)^1 (0.9)^{11} + 0.2001 \\ &= 1 \times 0.2824 + 12(0.1)(0.2138) + 0.2001 \\ &= \underline{0.8891} \end{aligned}$$

$$\begin{aligned} P(x \geq 2) &= 1 - P(x \leq 2) \\ &= 1 - [P(x=0) + P(x=1)] \\ &= 1 - 0.6899 \\ &= \underline{0.3101} \end{aligned}$$

$$v) P(x=0) = \underline{0.2824}$$

POISSON DISTRIBUTION.

Poisson distribution is regarded as the limiting case of the binomial distribution when 'n' is very large (i.e. $n \rightarrow \infty$) & the probability of success is very small ($p \rightarrow 0$) so that np tends to a fixed constant say ' λ '

find poisson distribution as a limiting case of the Binomial distribution

WKT the B.D is $P(x) = {}^nC_x p^x q^{n-x}$

$$\therefore P(x) = \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

the chance of occurrence of an event is very rare

it has a vast opportunity to occur in such situations are considered as P.D.

$$= \frac{n(n-1)(n-2) \dots (n-(x-1)) (n-x)!}{x!(n-x)!} p^x q^{n-x}$$

$$= \frac{n(n-1)(n-2) \dots (n-(x-1))}{x!} p^x q^{n-x}$$

$$= \frac{n(n-p)(n-2p) \dots (n-(x-1)p)}{x!} q^{n-x}$$

$$= \frac{\lambda(\lambda-p)(\lambda-2p) \dots (\lambda-(x-1)p)}{x!} q^{n-x}$$

taking limit as $n \rightarrow \infty$ & $p \rightarrow 0$, we get

$$P(x) = \frac{\lambda \cdot \lambda \cdot \lambda \dots \lambda}{x!} \lim_{n \rightarrow \infty} q^{n-x}$$

$$= \frac{\lambda^x}{x!} \lim_{n \rightarrow \infty} \frac{q^n}{q^x}$$

$$= \frac{\lambda^x}{x!} \lim_{n \rightarrow \infty} \frac{(1-p)^n}{(1-p)^x}$$

$$= \frac{\lambda^x}{x!} \lim_{n \rightarrow \infty} \frac{(1 - \lambda/n)^n}{(1 - \lambda/n)^x}$$

$$= \frac{\lambda^x}{x!} \frac{e^{-\lambda}}{1} \quad \left[\because \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x \right]$$

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

* S.T. mean & Variance of Poisson dist. are same
Proof:-

$$\text{Mean } \mu = \sum_{x=0}^{\infty} x p(x)$$

$$= \sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \sum_{x=1}^{\infty} \frac{e^{-\lambda} \lambda^x}{(x-1)!}$$

$$= \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!}$$

$$= \lambda e^{-\lambda} \left[1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \dots \right]$$

$$= \lambda e^{-\lambda} e^{\lambda}$$

$$\text{Mean} = \boxed{\mu = \lambda}$$

$$\text{Variance } V(x) = \sum_{x=0}^{\infty} x^2 p(x) - \mu^2 = E(x^2) - [E(x)]^2$$

$$= \sum_{x=0}^{\infty} x^2 \left(\frac{e^{-\lambda} \lambda^x}{x!} \right) - \lambda^2$$

$$= \sum_{x=0}^{\infty} \frac{(x(x-1) + x) e^{-\lambda} \lambda^x}{x!} - \lambda^2$$

$$= \sum_{x=0}^{\infty} x(x-1) \frac{e^{-\lambda} \lambda^x}{x!} + \sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!} - \lambda^2$$

$$\begin{aligned}
 &= \sum_{x=2}^{\infty} \frac{e^{-\lambda} \lambda^{x-2} \cdot \lambda^2}{(x-2)!} + \lambda - \lambda^2 \\
 &= e^{-\lambda} \lambda^2 \sum_{x=2}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} + \lambda - \lambda^2 \\
 &= e^{-\lambda} \lambda^2 e^{\lambda} + \lambda - \lambda^2 \\
 &= \lambda^2 + \lambda - \lambda^2
 \end{aligned}$$

Variance = $V(x) = \lambda$

Hence mean & Variance of Poisson distribution are same.

① The probabilities of a Poisson Variate taking the values 3 & 4 are equal. Calculate the Probabilities of the Variate taking the values 0 & 1.

Soln:- $P(x) = \frac{\lambda^x e^{-\lambda}}{x!}$

G.T $P(3) = P(4)$

$$\Rightarrow \frac{\lambda^3 e^{-\lambda}}{3!} = \frac{\lambda^4 e^{-\lambda}}{4!}$$

$$\Rightarrow \frac{\lambda^3}{6} = \frac{\lambda^4}{24}$$

$$\Rightarrow 6\lambda = 24 \Rightarrow \boxed{\lambda = 4}$$

$$\therefore P(x) = \frac{e^{-4} \cdot 4^x}{x!}$$

$$P(0) = e^{-4} = 1/e^4$$

$$\& P(1) = e^{-4} \cdot 4 = 4/e^4$$

(2) 2% of the fuses manufactured by a factory are found to be defective. Find the probability that a box containing 200 fuses contains
(i) no defective fuses (ii) 3 or more defective fuses

Soln:- P = prob. of a defective fuse

$$\therefore P = \frac{2}{100} = 0.02$$

Mean no of defectives $\mu = \lambda = np$

$$\Rightarrow \lambda = 200 \times 0.02$$

$$\Rightarrow \boxed{\lambda = 4}$$

$$P(x) = \frac{\lambda^x e^{-\lambda}}{x!} = \frac{4^x e^{-4}}{x!} = \frac{4^x (0.0183)}{x!}$$

(i) Prob. of no of defective fuse = $P(0) = 0.0183$

(ii) Prob. of 3 or more defective fuse = $P(x \geq 3)$

$$1 - P(x < 3) = 1 - [P(0) + P(1) + P(2)]$$

$$= 1 - 0.0183 \left[\frac{4^0}{0!} + \frac{4^1}{1!} + \frac{4^2}{2!} \right]$$

$$= \underline{0.7621}$$

(3) In a certain factory turning out razor blades there is small probability of $\frac{1}{500}$ for any blade to be defective. The blades are supplied in packets of 10, use poisson distribution to calculate the approximate no of packets containing (i) No defective (ii) one defective.

(iii) Two defective blades in a consignment of 10,000
Packets

Soln:-

$$P = \text{prob. of defective blade} = \frac{1}{500} = 0.002$$

In a packet of 10, the mean no. of defective blade is

$$\lambda = np = 10 \times 0.002 = 0.02$$

Poisson distribution is $p(x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-0.02} (0.02)^x}{x!}$

let $f(x) = 10,000 p(x)$

$$(e^{-0.02} = 0.9802)$$

$$f(x) = \frac{9802 (0.02)^x}{x!}$$

(i) Prob. of no defective $f(0) = 9802$

(ii) Prob. of one defective $f(1) = 9802 (0.02)$
 $= 196$

(iii) Prob. of two defectives $f(2) = \frac{9802 (0.02)^2}{2!}$
 $f(2) = 2$

- 4) A communication channel receives independent Pulse at the rate of 12 pulses per micro second. The probability of transmission error is 0.001 for each micro second. Compute the prob. of
- (i) no error during a micro second
 - (ii) one error per micro second
 - (iii) atleast one error per micro second
 - (iv) two errors
 - (v) atmost two errors

Soln:- Mean no of errors in one microsecond
is $\lambda = \mu = np = 12 \times 0.001 = 0.012$

The poisson distribution is $p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$

$$\text{i.e } p(x) = \frac{e^{-0.012} (0.012)^x}{x!}$$

$$(i) P(X=0) = 0.9881$$

$$(ii) P(X=1) = e^{-0.012} \times 0.012 = 0.01186$$

$$(iii) \text{ Prob. of atleast one error} = 1 - P(X=0) \\ = 1 - e^{-0.012} \\ = \underline{\underline{0.01193}}$$

$$(iv) P(X=2) = \frac{e^{-0.012} (0.012)^2}{2!} = 0.00071$$

$$(v) \text{ Prob. of at most two errors} \\ = P(X=0) + P(X=1) + P(X=2) \\ = e^{-0.012} \left[1 + 0.012 + \frac{(0.012)^2}{2} \right] \\ = \underline{\underline{0.9999 \approx 1}}$$

A car hire firm has two cars which it hires out every day. The no of demands for a car on each day is distributed as a Poisson distribution with mean 1.5. Calculate the proportion of days (i) on which there is no demand (ii) on which demands are refused

Ans: $\lambda = \text{mean demand for a car} = 1.5$

$$\therefore P(x) = \frac{e^{-1.5} (1.5)^x}{x!}$$

(i) No demand for a car. $P(0) = \frac{e^{-1.5} (1.5)^0}{0!} = 0.2231$

(ii) If the demand had to be refused, there should have been a demand for more than 2 cars.

$$\begin{aligned}\therefore P(x \geq 2) &= 1 - P(x \leq 1) \\ &= 1 - P(x=0) - P(x=1) \\ &= 1 -\end{aligned}$$

$$\underline{\underline{= 0.1913}}$$

Exponential Distribution

The continuous probability distribution having the Pdf. $f(x)$ given by

$$f(x) = \begin{cases} \alpha e^{-\alpha x}, & \text{for } x > 0 \\ 0, & \text{otherwise, where } \alpha > 0. \end{cases}$$

is known as the Exponential distribution

NOTE:-

$f(x) > 0$, we have.

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_0^{\infty} \alpha e^{-\alpha x} dx \\ &= \alpha \left[\frac{e^{-\alpha x}}{-\alpha} \right]_0^{\infty} = -(0-1) = 1 \end{aligned}$$

$$\therefore \int_{-\infty}^{\infty} f(x) dx = 1$$

$f(x)$ satisfies both the condⁿs required for a continuous probability fⁿ @ P.d.f.

S.I Mean & S.D of Exponential distribution are equal

Proof:- Mean (μ) = $\int_{-\infty}^{\infty} x \cdot f(x) dx$

$$= \int_0^{\infty} x \cdot \alpha e^{-\alpha x} dx$$

$$= \alpha \int_0^{\infty} x e^{-\alpha x} dx$$

$$= \alpha \left[x \frac{e^{-\alpha x}}{-\alpha} - (1) \frac{e^{-\alpha x}}{\alpha^2} \right]_0^{\infty}$$

$$= \alpha \left[0 - \frac{1}{\alpha^2} (0-1) \right]$$

$$\boxed{\mu = \frac{1}{\alpha}}$$

$$\text{Variance } (\sigma^2) = \int_{-\infty}^{\infty} (x-\mu)^2 f(x) dx$$

$$= \alpha \int_0^{\infty} (x-\mu)^2 \bar{e}^{-\alpha x} dx$$

$$\sigma^2 = \alpha \left[(x-\mu)^2 \frac{\bar{e}^{-\alpha x}}{-\alpha} - \frac{2}{\alpha^2} (x-\mu) \bar{e}^{-\alpha x} + 2 \frac{\bar{e}^{-\alpha x}}{-\alpha^3} \right]_0^{\infty}$$

$$= \alpha \left[0 - 0 + 0 - \left[\mu^2 \cdot \frac{1}{-\alpha} - 2(-\mu) \frac{1}{\alpha^2} - \frac{2}{\alpha^3} \right] \right]$$

$$\sigma^2 = \alpha \left[\frac{\mu^2}{\alpha} - \frac{2\mu}{\alpha^2} + \frac{2}{\alpha^3} \right]$$

$$\sigma^2 = \alpha \left[\frac{1}{\alpha^3} - \frac{2}{\alpha^3} + \frac{2}{\alpha^3} \right]$$

$$\sigma^2 = \frac{1}{\alpha^2}$$

$$\therefore \boxed{\sigma = \text{S.D.} = \frac{1}{\alpha}}$$

① If x is an exponential variate with mean 3 find (i) $P(x > 1)$ (ii) $P(x < 1)$

Soln:- The pdf of exponential distribution is given by

$$f(x) = \begin{cases} \alpha \bar{e}^{-\alpha x}, & 0 < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

$$\text{i.e. } f(x) = \alpha \bar{e}^{-\alpha x} \quad 0 < x < \infty$$

$$\text{Mean } \mu = \frac{1}{\alpha} = 3 \quad \text{i.e. } \boxed{\alpha = 1/3}$$

$$\text{Hence } f(x) = \frac{1}{3} e^{-x/3}, \quad 0 < x < \infty$$

$$) P(x > 1) = 1 - P(x \leq 1)$$

$$\int_1^{\infty} f(x) dx = 1 - \int_0^1 f(x) dx = 1 - \int_0^1 \frac{1}{3} \bar{e}^{-x/3} dx$$

$$= 1 + \left[e^{-x/5} \right]_0^1$$

$$P(x > 1) = \underline{e^{-1/5} = 0.7165}$$

$$\begin{aligned} \text{(ii)} \quad P(x < 1) &= \int_0^1 f(x) dx \\ &= \int_0^1 \frac{1}{5} e^{-x/5} dx = - \left[e^{-x/5} \right]_0^1 = 1 - \frac{1}{e} \end{aligned}$$

$$P(x < 1) = \underline{0.6321}$$

(2) In a certain town the duration of the shower is exponentially distributed with mean 5 min. What is the probability that a shower will last for (i) 10 min or more (ii) < 10 min (iii) between 10 & 12 min

Soln:-

$$\lambda = \frac{1}{\alpha} = 5 \Rightarrow \alpha = \frac{1}{5}$$

$$\therefore f(x) = \frac{1}{5} e^{-x/5}$$

$$\text{(i)} \quad P(x \geq 10) = \int_{10}^{\infty} \frac{1}{5} e^{-x/5} dx = \frac{1}{5} \left[\frac{e^{-x/5}}{-1/5} \right]_{10}^{\infty} = - (0 - 1) = \underline{0.1353}$$

$$\text{(ii)} \quad P(x < 10) = \int_0^{10} \frac{1}{5} e^{-x/5} dx = \frac{1}{5} \left[\frac{e^{-x/5}}{-1/5} \right]_0^{10} = 1 - e^{-2} = \underline{0.8647}$$

$$\text{(iii)} \quad P(10 < x < 12) = \int_{10}^{12} \frac{1}{5} e^{-x/5} dx$$

$$= \underline{0.0446}$$

(3) The length of telephone conversation in booth has been an exponential distribution & found on average to be 5 minutes. Find the probability that a random call made to this booth (i) ends less than 5 min.
(ii) Between 5 & 10 min

Soln. We have $f(x) = \lambda e^{-\lambda x}$, $x > 0$

$$\text{Mean } \mu = 1/\lambda$$

$$1/\lambda = 5$$

$$\Rightarrow \boxed{\lambda = 1/5}$$

$$\therefore f(x) = \frac{1}{5} e^{-x/5}$$

$$(i) P(x < 5) = \int_0^5 f(x) dx$$

$$= \int_0^5 \frac{1}{5} e^{-x/5} dx = -e^{-x/5} \Big|_0^5 = 1 - \frac{1}{e} = 0.61$$

$$(ii) P(5 \leq x \leq 10) = \int_5^{10} f(x) dx$$

$$= \int_5^{10} \frac{1}{5} e^{-x/5} dx = -e^{-x/5} \Big|_5^{10} = \frac{1}{e} - \frac{1}{e^2} = 0.21$$

(4) The sales per day in a shop is exponentially distributed with average sale amounting Rs. 100 & net profit is 8%. Find the probability that the net profit exceeds Rs. 20 on two consecutive days

Soln. let x be the random variable of sale in the shop

$$f(x) = \lambda e^{-\lambda x}, x > 0$$

$$\text{Mean} = \frac{1}{\lambda} = 100 \Rightarrow \boxed{\lambda = 0.01}$$

$$\therefore f(x) = 0.01 e^{-0.01x}, x > 0$$

let 'A' be the ~~turnover~~ ^{amount} for which the profit ~~is~~ net profit is Rs. 30 at the rate of 8%

$$\Rightarrow A \cdot \frac{8}{100} = 30 \quad \therefore \underline{A = 375}$$

Prob. of profit exceeding Rs. 30 is equal to

$$1 - \text{prob.}(\text{Profit} \leq 30) =$$

$$\Rightarrow 1 - \text{prob}(\text{Sales} \leq \text{Rs. } 375) \quad P(\text{Sales} > 375) = \int_{375}^{\infty} (0.01) e^{-0.01x} dx$$

$$\Rightarrow 1 - \int_0^{375} (0.01) e^{-0.01x} dx = 1 + \left[e^{-0.01x} \right]_0^{375} = e^{-3.75}$$

i.e. prob. that profit exceeds Rs. 30 on a single day is $e^{-3.75}$

$$\Rightarrow e^{-3.75} \cdot e^{-3.75} = e^{-7.5} = 0.00055$$

$$\text{Prob}(\text{profit} \geq 30)$$

$$\Rightarrow \text{Prob}(\text{Sale} \geq 375)$$

$$\Rightarrow \int_{375}^{\infty} (0.01) e^{-0.01x} dx = 0.01 \left[\frac{e^{-0.01x}}{-0.01} \right]_{375}^{\infty}$$

$$\Rightarrow \underline{e^{-3.75}}$$

(6) $x \rightarrow$ R.v denoting the increase of sales per day
 $f(x) = x e^{-\frac{1}{600}x} = \frac{1}{600} e^{-\frac{1}{600}x}$

let A be the sales for which the sales tax is Rs. 81 at the rate of 9%. $\Rightarrow A \cdot \frac{9}{100} = 81$

$$P(x > 900) = \int_{900}^{\infty} \frac{1}{600} e^{-\frac{1}{600}x} dx = \frac{1}{600} \left(\frac{e^{-0.0016x}}{-0.0016} \right)_{900}^{\infty}$$

$$P(x > 900)$$

Uniform distribution

The continuous p.d with pdf is given by

$$f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}, \text{ where } a \text{ \& } b \text{ are arbitrary real constants}$$

is called Uniform [rectangular] probability distribution

Here $f(x) \geq 0$ & $\int_{-\infty}^{\infty} f(x) dx = 1$

Mean

$$\mu = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_a^b x \cdot \frac{1}{b-a} dx = \frac{1}{(b-a)} \left[\frac{x^2}{2} \right]_a^b = \frac{1}{2(b-a)} (b^2 - a^2)$$

$$\boxed{\mu = \frac{(b+a)}{2}}$$

variance

$$V(x) = \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2$$

$$= \int_a^b x^2 \frac{1}{(b-a)} dx - \left(\frac{b+a}{2} \right)^2$$

$$= \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b - \frac{(b+a)^2}{4}$$

$$= \frac{1}{b-a} \left[\frac{b^3 - a^3}{3} \right] - \frac{(b+a)^2}{4}$$

$$= \frac{1}{3(b-a)} [(b-a)(b^2 + ab + a^2)] - \frac{(b+a)^2}{4}$$

$$= \frac{4b^2 + 4ab + 4a^2 - 3b^2 - 3a^2 - 6ab}{12}$$

$$= \frac{b^2 + a^2 - 2ab}{12} = \frac{(b-a)^2}{12} \therefore V(x) = \frac{(b-a)^2}{12}$$

S.D.

$$\sigma = \frac{b-a}{2\sqrt{3}}$$

① A random Variable X is uniform distributed over $(-2, 2)$. find

(i) $P(X < 1)$

(ii) $P(|X-1|)$

(iii) σ (iv)

Soln:- G.T $a = -2, b = 2$

$$f(x) = \begin{cases} \frac{1}{4}, & -2 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

$$\therefore \mu = \frac{b+a}{2} = \frac{2-2}{2} = 0, \quad \sigma = \frac{1}{2\sqrt{3}} \cdot [2+2]$$

$$(i) P(X < 1) = \int_{-2}^1 f(x) dx = \int_{-2}^1 \frac{1}{4} dx = \frac{3}{4}$$

$$(ii) P(|X| > 1) = 1 - P(|X| \leq 1) \\ = 1 - \int_{-1}^1 \frac{1}{4} dx = \underline{\underline{\frac{1}{2}}}$$

$$(iii) P(|X-1| \geq \frac{1}{2}) = 1 - P(|X-1| < \frac{1}{2}) \\ = 1 - P\left(-\frac{1}{2} < (X-1) < \frac{1}{2}\right) \\ = 1 - P\left(\frac{1}{2} < X < \frac{3}{2}\right) \\ = 1 - \int_{1/2}^{3/2} \frac{1}{4} dx \\ = 1 - \left(\frac{3}{2} - \frac{1}{2}\right) \\ = \underline{\underline{1}}$$

Q) On a certain city transport route, buses ply every 30 minutes b/w 6 a.m & 10 p.m. If a person reaches a bus stop on this route at a random time during this period, what is the probability that he will have to wait for at least twenty minutes?

Soln:- let x be the waiting time (in min) for the next bus.

let x is distributed uniformly over the interval $(0, 30)$ with pdf

$$\therefore f(x) = \begin{cases} \frac{1}{30}, & 0 < x < 30 \\ 0, & \text{otherwise} \end{cases}$$

$$\therefore P(x \geq 20) = \int_{20}^{30} \frac{1}{30} dx = \frac{1}{30} [x]_{20}^{30} = \frac{1}{30} [30 - 20] = \frac{1}{3}$$

Q) If the Random Variable k is uniformly distributed over $(0, 5)$. What is the prob. that the roots of the eqn $4x^2 + 4kx +$

$(k+2) = 0$ are real

Soln:- $f(k) = \begin{cases} \frac{1}{5-0}, & 0 \leq k \leq 5 \\ 0, & \text{otherwise} \end{cases}$

$$a=0 \\ b=5$$

$$4x^2 + 4kx + (k+2) = 0.$$

$$x = \frac{-4k \pm \sqrt{(4k)^2 - 4(4)(k+2)}}{2(4)} = \frac{-k \pm \sqrt{k^2 - k - 2}}{2}$$

$$x = \frac{-k}{2} \pm \frac{\sqrt{k^2 - (k+2)}}{4}$$

$$\text{Discriminant} = k^2 - (k+2)$$

$$P[\text{roots of the eqn } 4x^2 + 4kx + (k+2) = 0 \text{ are real}] = P[\text{discriminant of the eqn} \geq 0]$$

$$= P[(k^2 - (k+2)) \geq 0]$$

$$= P[k^2 - k - 2 \geq 0]$$

$$= P[(k+1)(k-2) \geq 0]$$

$$= P[(k+1) \geq 0, (k-2) \geq 0]$$

$$= P[k \geq -1, k \geq 2]$$

$$= P[k \geq 2] \left[\because k \geq -1 \text{ doesn't lead in (0.5)} \right]$$

$$= P[2 \leq k \leq \infty]$$

$$= \int_2^{\infty} f(k) dk$$

$$= \int_2^5 f(k) dk + \int_5^{\infty} f(k) dk$$

$$= \int_2^5 \frac{1}{5} dk$$

$$= \frac{1}{5} [5-2]$$

$$= \frac{3}{5} = 0.6$$

Normal distribution (D1)

Gaussian distribution

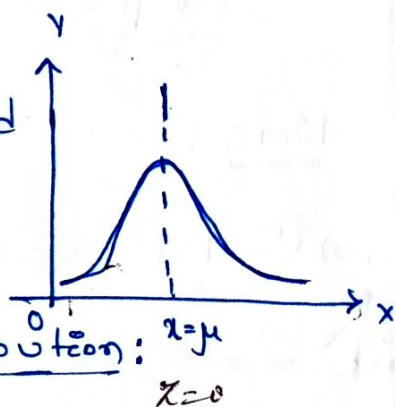
The continuous probability distribution function having pdf $f(x)$ given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad \text{where } -\infty < x < \infty$$

$-\infty < \mu < \infty$ & $\sigma > 0$ is known as Normal distribution

clearly $f(x) \geq 0$ & $\int_{-\infty}^{\infty} f(x) dx = 1$

The graph of $f(x)$ is a bell shaped curve symmetrical about $x = \mu$ (mean).



Standard form of Normal distribution: $x = \mu$
 $x = 0$

$$P(a \leq x \leq b) = \frac{1}{\sigma\sqrt{2\pi}} \int_{x=a}^b e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}} dx \rightarrow (1)$$

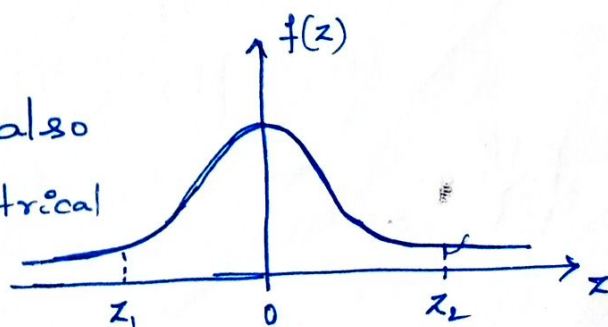
To evaluate the above integral, we need to standardize the function, by putting $z = \frac{x-\mu}{\sigma}$ is called Standard normal Variate

The (1) $\Rightarrow P(a \leq x \leq b) = \frac{1}{\sqrt{2\pi}} \int_{z_1}^{z_2} e^{-\frac{z^2}{2}} dz = P(z_1 \leq z \leq z_2)$ where $z_1 = \frac{a-\mu}{\sigma}$
 $z_2 = \frac{b-\mu}{\sigma}$
 $\rightarrow (2)$
 $dx = \sigma dz$

then $f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$ is called Standard normal distribution function

which is same as the pdf of the normal distribution with $\mu = 0$ & $\sigma = 1$

Hence the graph of $f(z)$ is also a bell shaped curve symmetrical about the line $z = 0$.



① The integral $\phi(z) = \frac{1}{\sqrt{2\pi}} \int_{z_1}^{z_2} e^{-z^2/2} dz = A(z)$

represents the area bounded by the Normal Curve $f(z)$ b/w $z=z_1$ & $z=z_2$

② Since the total area is 1, the area on either side of $z=0$ is 0.5.

$$\int_{-\infty}^{\infty} f(z) dz = 1, \quad \int_{-\infty}^0 f(z) dz = 0.5 = \int_0^{\infty} f(z) dz$$

NOTE :

① $P(-\infty \leq z \leq \infty) = 1$

② $P(-\infty \leq z \leq 0) = \frac{1}{2}$ & $P(0 \leq z \leq \infty) = P(z \geq 0) = \frac{1}{2}$

③ $P(-\infty < z \leq z_1) = P(-\infty < z \leq 0) + P(0 \leq z \leq z_1)$

③ $P(z_1 \leq z \leq \infty) = P(z \geq 0) - P(0 \leq z \leq z_1)$
 $= 0.5 - \phi(z_1)$

(1) $P(-\infty \leq z \leq \infty) = 1$

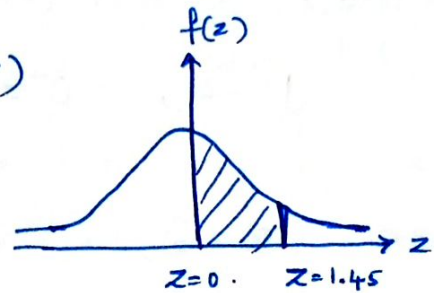
(2) $P(-\infty \leq z \leq 0) = P(0 \leq z \leq \infty) = P(z \geq 0) = 0.5$

(3) If $z_1 \geq 0$, $P(z_1 \leq z \leq \infty) = P(z \geq 0) - P(0 \leq z \leq z_1)$
 $= 0.5 - \phi(z_1)$

① for the Std normal distribution of a R.V Z evaluate

(i) $P(0 \leq Z \leq 1.45) = P(Z \leq 1.45) - P(Z \leq 0) = \phi(1.45)$

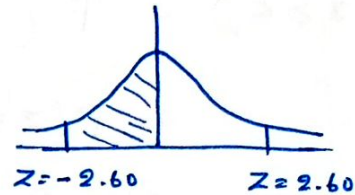
$= \underline{0.4265}$



(ii) $P(-2.60 \leq Z \leq 0) = P(0 \leq Z \leq 2.60)$

$= \phi(2.60)$

$= \underline{0.4953}$



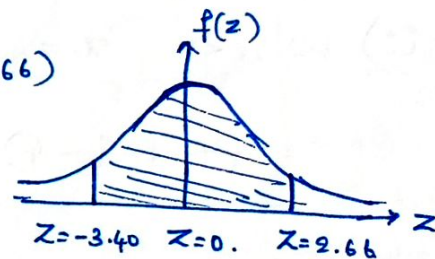
(iii) $P(-3.40 \leq Z \leq 2.66) = P(-3.40 \leq Z \leq 0) + P(0 \leq Z \leq 2.66)$

$= P(0 \leq Z \leq 3.40) + \phi(2.66)$

$= \phi(3.40) + \phi(2.66)$

$= 0.4997 + 0.4961$

$= \underline{0.9958}$

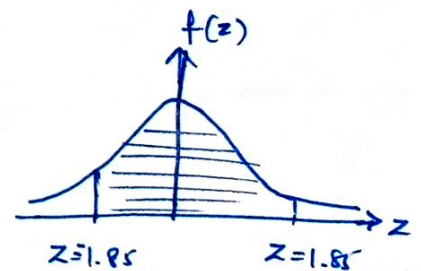


(iv) $P(|Z| \leq 1.85) = P(-1.85 \leq Z \leq 1.85)$

$= 2\phi(1.85)$

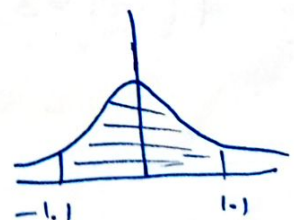
$= 2 \times 0.4678$

$= \underline{0.9356}$



(v) $P(-1.1 \leq Z \leq 1.1) = 2\phi(1.1)$

$= 2 \times 0.3643 = \underline{0.7286}$

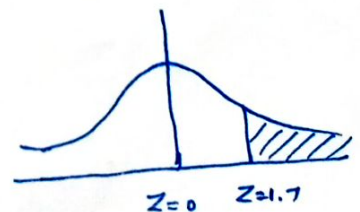


(vi) $P(Z \geq 1.7) = P(Z \geq 0) - P(0 \leq Z \leq 1.7)$

$= 0.5 - \phi(1.7)$

$= 0.5 - 0.4554$

$= \underline{0.0446}$



② For the normal distribution with $\mu = 2$ & S.D. 4, evaluate the following probabilities

(i) $P(x \geq 5)$ (ii) $P(|x| < 4)$ (iii) $P(|x| > 3)$

Soln: $\mu = 2$, $\sigma = 4$, $Z = \frac{x - \mu}{\sigma}$ be the St. Var

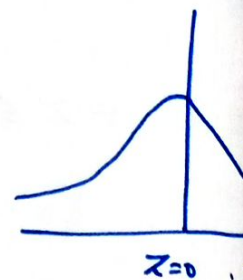
(i) when $x = 5$, $Z = 0.75$

$$\therefore P(x \geq 5) = P(Z \geq 0.75)$$

$$= P(Z \geq 0) - P(Z \leq 0.75)$$

$$= 0.5 - 0.2734$$

$$= \underline{0.2266}$$



(ii) when $x = 4 \Rightarrow Z = 0.5$

$$P(|x| < 4) \Rightarrow P(|Z| \leq 0.5)$$

$$= P(-0.5 < Z < 0.5)$$

$$= 2\phi(0.5)$$

$$= 2 \times 0.1915$$

$$= \underline{0.383}$$

(iii) when $x = 3 \Rightarrow Z = 0.25$

$$P(|x| > 3) \Rightarrow P(|Z| > 0.25)$$

$$\Rightarrow 1 - P(|Z| \leq 0.25)$$

$$\Rightarrow 1 - 2\phi(0.25)$$

$$\Rightarrow 1 - 2[0.0987]$$

$$\Rightarrow \underline{0.8026}$$

- ③ The marks of 1000 students in an examination follows a normal distribution with mean 70 & S.D. 5. find the no of students whose marks will be (i) less than 65 (ii) more than 75 & (iii) b/w 65 & 75

x be the marks scored by the students

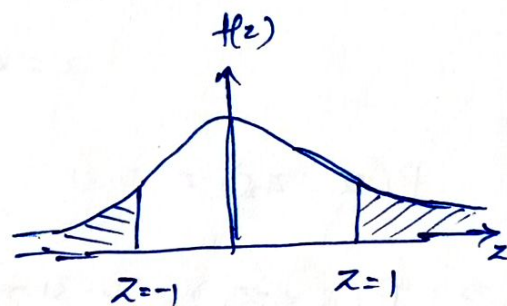
Soln. $\mu = 70, \sigma = 5$

$$\therefore Z = \frac{x - 70}{5}$$

(i) when $x = 65, Z = -1$

$$\begin{aligned} \therefore P(x < 65) &= P(Z < -1) \\ &= P(Z \geq 1) \\ &= P(Z \geq 0) - P(0 < Z < 1) \\ &= 0.5 - \phi(1) \\ &= 0.5 - 0.3413 \end{aligned}$$

$$P(x < 65) = 0.1587$$



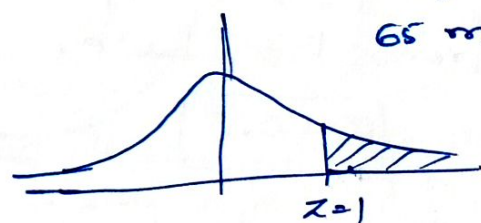
for 1000 students

$$1000 \times 0.1587 = 158.7$$

159 Students scored less than 65 marks

(ii) when $x = 75, Z = 1$

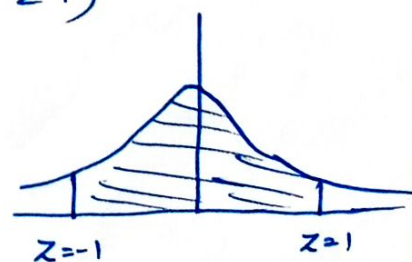
$$\begin{aligned} P(x > 75) &= P(Z > 1) \\ &= 0.5 - \phi(1) \\ &= 0.1587 \end{aligned}$$



$\therefore$ for 1000 students $1000 \times 0.1587 = 158.7 \approx 159$ students scored ~~less~~ ^{more} than 75 marks

(iii) when $P(65 < x < 75) = P(-1 < Z < 1)$

$$\begin{aligned} P(65 < x < 75) &= 2\phi(1) \\ &= 2 \times 0.3413 \\ &= 0.6826 \end{aligned}$$



for 1000 students $1000 \times 0.6826 = 682.6 \approx 683$ students scored marks b/w 65 & 75

④ In a normal distribution, 31% of items are under 45 & 8% are over 64, find the mean & standard deviation of the distribution

Soln:-

$$Z = \frac{x - \mu}{\sigma}$$

When $x = 45$, let $Z = z_1$, $\therefore -z_1 = \frac{45 - \mu}{\sigma} \rightarrow$

$x = 64$, let $Z = z_2$, $\therefore z_2 = \frac{64 - \mu}{\sigma} \rightarrow$

$$P(x < 45) = 0.31$$

$$\Rightarrow P(Z < -z_1) = 0.31$$

(where z_1 be negative)

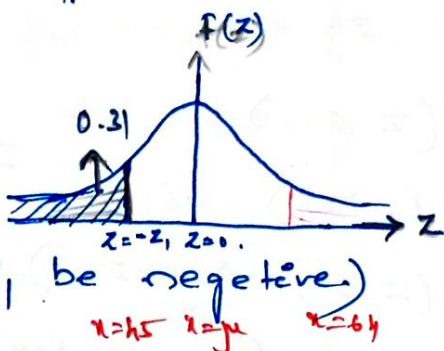
$$P(Z > z_1) = 0.31$$

$$0.5 - \phi(z_1) = 0.31$$

$$\phi(z_1) = 0.19$$

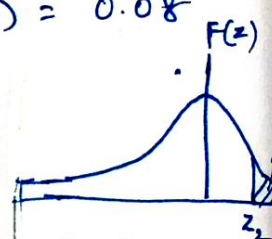
from the table.

$$\boxed{z_1 = 0.5}$$



$$P(x > 64) = 0.08$$

$$P(Z > z_2) = 0.08$$



$$\Rightarrow 0.5 - \phi(z_2) = 0.08$$

$$\Rightarrow \phi(z_2) = 0.42$$

from the table.

$$\boxed{z_2 = 1.4}$$

Use in ① & ②

$$\Rightarrow -0.5 = \frac{45 - \mu}{\sigma}, \quad 1.4 = \frac{64 - \mu}{\sigma}$$

$$\Rightarrow -0.5\sigma + \mu = 45, \quad 1.4\sigma + \mu = 64$$

Solving these, we get

$$\boxed{\mu = 50}$$

$$\boxed{\sigma = 10}$$

Steel rods are manufactured to be 3 cms in diameter but they are acceptable if they are inside the limit 2.99 cms & 3.01 cms. It is observed that 5% are rejected as oversized & 5% are rejected as undersized. Assuming that the diameters are normally distributed, find the S.D of the distribution.

∴ Given $\mu = 3 \text{ cms}$, $\sigma = ?$

$$Z = \frac{x - 3}{\sigma}$$

When $x = 2.99$, $Z = Z_1 = \frac{2.99 - 3}{\sigma} = -\frac{0.01}{\sigma} = Z_1$

$x = 3.01$, $Z = Z_2 = \frac{3.01 - 3}{\sigma} = \frac{0.01}{\sigma} = Z_2$

$P(x < 2.99) = 0.05$, $P(x > 3.01) = 0.05$

$P(Z < Z_1) = 0.05$

$P(Z < -\frac{0.01}{\sigma}) = 0.05$

$P(Z > \frac{0.01}{\sigma}) = 0.05$

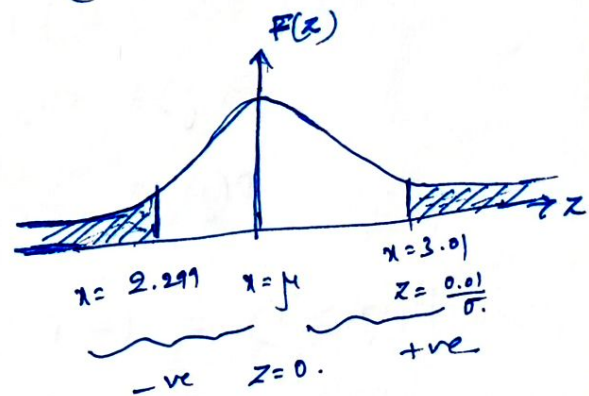
$P(Z \geq 0) - P(0 \leq Z \leq \frac{0.01}{\sigma}) = 0.05$

$0.5 - \Phi(\frac{0.01}{\sigma}) = 0.05$

$\Phi(\frac{0.01}{\sigma}) = 0.45$

$\therefore \frac{0.01}{\sigma} = 1.645$

$\therefore \sigma = 0.006$



$\therefore \Phi(1.645) = 0.45$

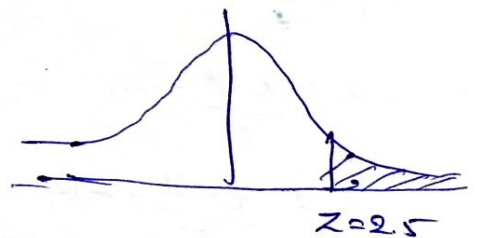
An analog signal received as a detector (measured in micro volts) may be modeled as a Gaussian R.v. (Normal) with mean 200 Variance 256 at a fixed point of time. What is the prob. that the signal will be 240 micro volts?

Soln:- Given $\mu = \text{mean} = 200$,
 $\sigma^2 = 256$
 $\sigma = \text{S.D} = 16$

$$Z = \frac{x - \mu}{\sigma} \Rightarrow Z = \frac{x - 200}{16}$$

When $x = 240$, $Z = \frac{240 - 200}{16} = 2.5$

$$\begin{aligned} P(x > 240) &= P(Z > 2.5) \\ &= 0.5 - \phi(2.5) \\ &= 0.5 - 0.4938 \\ &= \underline{\underline{0.0062}} \end{aligned}$$



The life span of an electric bulb is normally distributed with mean 2040 hrs & S.D 60 hrs. In a consignment of 3000 lamps, how many would be expected to burn for (i) more than 2150 hrs (ii) less than 1950 hrs & (iii) b/w 1920 & 2160 hrs.

$$\mu = 2040, \quad \sigma = 60$$

$x \rightarrow$ expected life of each bulb.

$$Z = \frac{x - \mu}{\sigma} = \frac{x - 2040}{60}$$

(i) when $x = 2150$, $Z = \frac{2150 - 2040}{60} = 1.83$

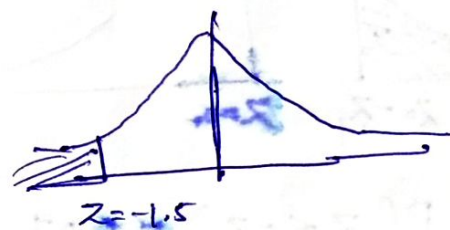
$$\begin{aligned} P(x > 2150) &= P(Z > 1.83) \\ &= 0.5 - \phi(1.83) \\ &= 0.5 - 0.4641 \\ &= \underline{0.0359} \end{aligned}$$



for 3000 lamps $\rightarrow 3000 \times 0.0359 = 107.7$
 $\approx$ 108 bulbs will burn more than 2150 hrs

(ii) $x = 1950 \Rightarrow Z = -1.5$

$$\begin{aligned} P(x < 1950) &= P(Z < -1.5) \\ &= P(Z > 1.5) \\ &= 0.5 - \phi(1.5) \\ &= 0.5 - 0.4332 \\ &= \underline{0.0668} \end{aligned}$$



for 3000 lamps $\rightarrow 3000 \times 0.0668 = 200.4 \approx$ 200
 bulbs will burn more than 1950 hrs

(iii) $x = 1920 \Rightarrow Z = -2$ $x = 2160 \Rightarrow Z = 2$

$$\begin{aligned} P(1920 < x < 2160) &= P(-2 < Z < 2) \\ &= 2\phi(2) \\ &= 2(0.4772) = 0.9544 \end{aligned}$$

for 3000 lamps $\rightarrow 3000 \times 0.9544 = 2863.2$
 $\approx$ 2863 bulbs
 will burn b/w 1920 & 2160 hrs

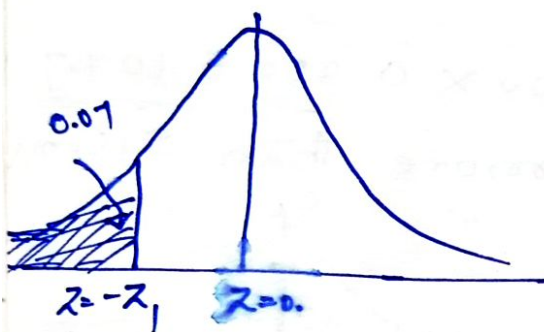
In a normal distribution 7% are under 35 & 89% are under 63. find the mean & the SD

Soln:- If $x = 35 \Rightarrow Z = \frac{x - \mu}{\sigma} = \frac{35 - \mu}{\sigma} = -Z_1 \rightarrow (1)$

$x = 63 \Rightarrow Z = \frac{63 - \mu}{\sigma} = Z_2 \rightarrow (2)$

$P(x < 35) = 0.07$

$P(x < 63) = 0.89$



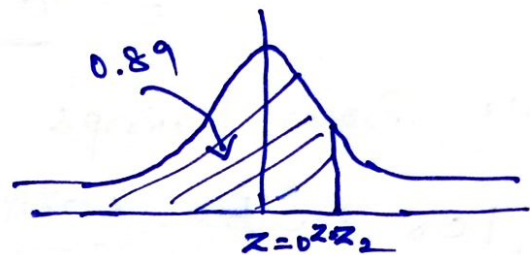
$P(Z < -Z_1) = 0.07$

$P(Z > Z_1) = 0.07$

$\phi(0.5) - \phi(Z_1) = 0.07$

$\phi(Z_1) = 0.43$

$Z_1 = 1.48$



$P(Z < Z_2) = 0.89$

$0.5 + \phi(Z_2) = 0.89$

$\phi(Z_2) = 0.89 - 0.5$

$\phi(Z_2) = 0.39$

$Z_2 = 1.23$

in (1) & (2) $\Rightarrow \frac{35 - \mu}{\sigma} = -1.48$

$\frac{63 - \mu}{\sigma} = 1.23$

$\Rightarrow 35 - \mu = -1.48\sigma$

$63 - \mu = 1.23\sigma$

$\Rightarrow \mu - 1.48\sigma = 35$

$\mu + 1.23\sigma = 63$

$\mu = 50.29$

$\sigma = 10.33$

⑥ The marks obtained by a no of students in a certain subject are approximately normally distributed with mean 65 & S.D 5. If 3 students are selected at random from this group. What is the prob. that at least one of them would have scored above 75

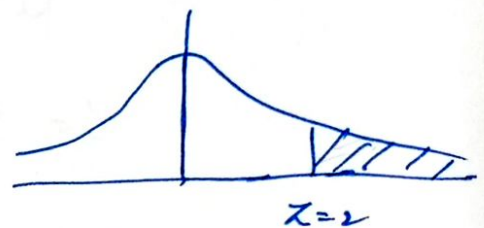
Soln:- $\mu = 65$, $\sigma = 5$. Let x be the marks scored by a student

$$Z = \frac{x - 65}{5} = \frac{75 - 65}{5}$$

$$P(x > 75) = P(Z > 2) = 0.5 - \phi(2)$$

$$P(x > 75) = 0.5 - 0.4772$$

$$P(x > 75) = 0.0228$$



$y \rightarrow$ no of students scored above 75 mark
 $n = 3$, $P = 0.0228$, $q = 0.9772$

$$P(y) = {}^nC_y P^y q^{n-y}$$

$$P(y) = {}^3C_y (0.0228)^y (0.9772)^{3-y}$$

$$P(y \geq 1) = 1 - P(y < 1)$$

$$= 1 - P(y = 0)$$

$$= 1 - {}^3C_0 (0.0228)^0 (0.9772)^3$$

$$= 1 - 0.933$$

=

$$P(y \geq 1) = 0.0668$$

1) In an engineering examination, a student considered to have failed, secured 2nd class & distinction, according as he scored less than 45%, b/w 45% & 60%, b/w 60% & 75% above 75% respectively. In a particular year 1% of the students failed in the examination, 5% of the students got distinction. Find the percentage of students who have got 1st class & 2nd class (Assume normal distribution)

- 01)-
- 1st class ^{fail} — $\leq 45\%$
 - 2nd class — $45 < x < 60$
 - 1st class — $60 < x < 75$
 - Distinction — > 75

let x be the normal variate gives the % of marks of the student

$$P(x < 45) = 0.01$$

$$Z = \frac{x - \mu}{\sigma} = -Z_1$$

$$Z_1 = \frac{45 - \mu}{\sigma} = -Z_1$$

$$P(Z < -Z_1) = 0.01$$

$$P(Z > Z_1) = 0.01$$

$$0.5 - \phi(Z_1) = 0.01$$

$$\phi(Z_1) = 0.49$$

$$Z_1 = 2.33$$

$$P(x \geq 75) = 0.05$$



$$Z_2 =$$

$$\sigma, \mu =$$